returns using a variant of Black-Litterman (1991). I set the risk-free rate to be 2%. I compute mean-variance frontiers constraining the sector weights to be positive.

We start with all the sectors. Then, I'll remove tobacco. Next I'll remove the aerospace and defense sector. Norway has selectively divested some companies in this sector because it automatically excludes all companies involved in the manufacture of nuclear weapons and cluster munitions.[38] A final exclusion that I'll examine is banks. Sharia law prohibits the active use of derivatives and debt as profit-making activities. Thus, it is interesting to see what diversification costs are routinely incurred by some Sharia compliant funds.

As we move from the full universe to the restricted universe, we obtain the following minimum standard deviations and maximum Sharpe ratios:

	Minimum Volatility	**Maximum Sharpe Ratio**
All Sectors	0.1205	0.4853
No Tobacco	0.1210	0.4852
No Tobacco and Aerospace & Defense	0.1210	0.4852
No Tobacco, Aerospace & Defense, and Banks	0.1210	0.4843

The increase in the minimum volatility is tiny—from 12.05% to 12.10%. Similarly, the reduction in maximum Sharpe ratio is negligible, moving from 0.4853 for the full universe to 0.4852 when tobacco is removed and to 0.4843 when all three sectors are removed. Thus, the loss in diversification from removing one or a few sectors is extremely small. Figure 3.17 plots the (constrained) mean-variance frontiers for each set of sectors. They are indistinguishable on the graph. Norway is effectively losing nothing by selling Wal-Mart.[39] It is also effectively losing nothing by excluding tobacco.

It is important to note that in this example that I am computing the loss of ex-ante diversification benefits, as measured by the minimum variance portfolio or the maximum Sharpe ratio portfolio—concepts that are relevant for an investor selecting from the FTSE sectors. I am not considering tracking error relative to the full FTSE universe or ex-post differences in returns (which is relevant for an investor forced to track the FTSE index).

[38] As of June 2012, nineteen defense manufacturers had been excluded. A full list of Norway's current exclusions is at http://www.regjeringen.no/en/dep/fin/Selected-topics/the-government-pension-fund/responsible-investments/companies-excluded-from-the-investment-u.html?id=447122.

[39] This does not include the actual transaction costs of divestment. My case "The Norwegian Government Pension Fund: The Divestiture of Wal-Mart Stores Inc.," Columbia CaseWorks, ID#080301, also estimates these transaction costs.